

NCKU – Causal Inference Syllabus

February 2025 ~ June 2025

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(戴安順, 成大統計系)

Couse outline

Part 1. Introduction

- 1-1. Introduction to Causal Inference
- 1-2. Counterfactual Framework and Causal Estimands
- 1-3. Assumption and Identification

Part 2. Randomized Experiments

- 2-1. Assignment mechanisms
- 2-2. Classical randomized experiments

Part 3. Observational Studies with Measured Confounding

- 3-1. Stratification via covariates
- 3-2. Standardization
- 3-3. Propensity-score methods

Part 4. Observational Studies with Unmeasured Confounding

- 4-1. Front-door criterion
- 4-1. Difference-in-Differences (DiD) method
- 4-2. Instrumental variable

Part 5. Sensitivity Analysis

- 5-1. Evaluating sensitivity to exchangeability assumption violations
- 5-2. Evaluating sensitivity to positivity assumption violations

Part 6. Causal Directed Acyclic Graphs (DAGs)

- 6-1. DAGs for selection bias and confounding bias
- 6-2. DAGs for measurement bias
- 6-3. DAGs for interaction and effect modification

Part 7. Causal Mediation Analysis

- 7-1. Product method and difference method
- 7-2. Mediation G-formula
- 7-3. Estimation for causal mediation analysis

Part 8. Causal Inference for Longitudinal Data

- 8-1 G-method for time-vary treatments and confounders
- 8-2 Censoring and truncation
- 8-3 Causal survival analysis and survival average causal effect

Prerequisites

- Basic knowledge of probability theory,
- Understanding of statistical inference,
- Familiarity with linear regression,
- Familiarity with logistic regression,
- Experience with R programming
- Basic knowledge of survival analysis (optional)

Textbook

This course has no specific textbook and is primarily based on lecture notes and various papers.

Reference book

- Ding, P. (2024). *A First Course in Causal Inference*.
- Hernán, M. A. & Robins, J. M. (2020). *Causal Inference: What If*.
- Brumback, B. A. (2021). *Fundamentals of Causal Inference: With R*.
- Pearl, J., & Mackenzie, D. (2018). *The book of why: the new science of cause and effect*.
Basic books.

Grading

Your grade will be determined by **Homework Assignments (50%) and Final Project (50%)**.